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Cambridge IGCSE Combined and Co-ordinated Sciences

C3 Stoichiometry

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Comprehensive Exam Preparation Guide

C3 — Stoichiometry

1. Chemical Formulae and Equations (C3.01)

What Is Stoichiometry?

Stoichiometry comes from the Greek words stoicheion (element) and metron (measure). It is "chemical accountancy" at the level of atoms: using formulae, balanced equations and the mole to work out exactly how much of each substance reacts and how much product is made.

- The **mole** is the chemist's standard amount of substance; it contains a known number of particles.
- Industry uses these calculations to run processes safely and economically, with as little waste as possible. This is the aim of the 12 principles of **Green Chemistry**, which include maximising **atom economy**.

Formulae of Elements

Every element has its own **chemical symbol**, shown in the Periodic Table:

- A single capital letter, e.g. hydrogen H, oxygen O, nitrogen N.
- Or two letters, e.g. aluminium Al, argon Ar, magnesium Mg. The **second letter is always lower case**.
- Some symbols come from the element's **Latin name**:

Element	Latin name	Chemical symbol
Silver	argentum	Ag
Gold	aurum	Au
Copper	cuprum	Cu
Sodium	natrium	Na
Potassium	kalium	K
Iron	ferrum	Fe
Lead	plumbum	Pb

The formula of an element depends on its structure:

Structure of the element	Formula	Examples
Single atoms (the noble gases)	Just the symbol	He, Ne, Ar, Kr
Giant metallic lattice (metals)	Just the symbol	Li, Na, K, Mg, Ca, Al, Cu, Zn, Fe
Giant covalent structure	Just the symbol	B, C, Si, Ge
Simple molecules	The symbol with a subscript showing the number of atoms in one molecule	H_2, N_2, O_2, F_2, Cl_2, Br_2, P_4, S_8

- Hydrogen, nitrogen, oxygen and the halogens are **diatomic**: each molecule contains two atoms, so they are always written H_2, N_2, O_2, Cl_2 in equations.

Formulae of Compound Ions

- **Simple ions** are single atoms that have lost or gained electrons, e.g. Na^+, K^+, Mg^{2+}, Cl^-, O^{2-}.
- A **compound ion** is a group of different atoms covalently bonded together that carries an overall charge. Most are **negative** because the group has gained electrons; negative compound ions containing oxygen are called **oxyanions**.
- **Ammonium, NH_4^+**, is the common **positive** compound ion.
- In diagrams a compound ion is drawn inside square brackets with the charge outside, because the charge is spread over the whole ion.
- The formula of a compound ion never changes: the carbonate ion is always CO_3^{2-}.

Compound ion	Formula	Atoms in the ion	Charge
Carbonate	CO_3^{2-}	1 C, 3 O	2−
Nitrate	NO_3^-	1 N, 3 O	1−
Sulfate	SO_4^{2-}	1 S, 4 O	2−
Hydrogencarbonate	HCO_3^-	1 H, 1 C, 3 O	1−
Hydroxide	OH^-	1 O, 1 H	1−
Ammonium	NH_4^+	1 N, 4 H	1+

Formulae of Simple Covalent Compounds

The combining power of an atom is sometimes called its **valency**: in a covalent molecule, it is the number of bonds that the atom makes. (The idea is useful for working out formulae, but the term valency itself is not required knowledge.)

- Groups I–IV: valency = group number.
- Groups V–VII: valency = 8 − group number.
- Group VIII: valency = 0 (the noble gases do not normally form compounds).

Group (Period 3)	Valency	Typical compound
I	1	NaCl
II	2	$MgCl_2$
III	3	$AlCl_3$
IV	4	$SiCl_4$
V	3	PH_3
VI	2	H_2S
VII	1	HCl
VIII	0	—

The cross-over method (Worked Example C3.01, carbon dioxide):

Step 1: Write down the symbols: C and O.

Step 2: Write the valency under each symbol: C = 4, O = 2.

Step 3: Cross over the valencies, so that each becomes the number of atoms of the other element: C_2O_4.

Step 4: Simplify to the lowest whole-number ratio: **CO_2**.

The method works for molecules with a central atom, e.g. water H_2O (H = 1, O = 2), methane CH_4 (C = 4, H = 1), carbon dioxide CO_2 and ammonia NH_3 (N = 3, H = 1).

Molecular formula from a diagram: count the atoms of each element in the structure. For example (Figure C3.05), one Br atom bonded to three F atoms is **BrF_3**; one N atom bonded to three H atoms is **NH_3**; the propene molecule $H_2C{=}CH{-}CH_3$ contains 3 C and 6 H atoms, so it is **C_3H_6**.

Formulae of Ionic Compounds

- Ionic compounds have **giant ionic lattice** structures, so their formula is simply the **whole-number ratio of positive to negative ions**. This group of ions is one **formula unit**.
- The overall structure must be **neutral**: the positive and negative charges must balance.
- Example: in magnesium chloride the ions present are Mg^{2+} and $2Cl^-$, with total charges of 2+ and 2−, so the formula is **$MgCl_2$**. Mg^{2+} ions combine with Cl^- ions in a 1 : 2 ratio, but Na^+ ions combine with Cl^- ions only in a 1 : 1 ratio (NaCl).

Working out the charge on an ion:

- Metals and hydrogen form **positive** ions; non-metals form **negative** ions.
- Groups I–IV: charge = group number (e.g. Na^+, Mg^{2+}, Al^{3+}).
- Groups V–VII: charge = 8 − group number (e.g. O^{2-}, Cl^-).
- Transition elements can have more than one charge; the Roman numeral in the name gives the charge, e.g. copper(I) is Cu^+, copper(II) is Cu^{2+} and iron(III) is Fe^{3+}.

Worked Example C3.02 (cross over the charges):

- Aluminium oxide: Al has a charge of 3+ and O has a charge of 2−. Cross over: **Al_2O_3** (2 × 3+ = 6+ balances 3 × 2− = 6−).
- Calcium oxide: Ca has a charge of 2+ and O has a charge of 2−. Cross over: Ca_2O_2, then simplify the ratio: **CaO**.

Compound ions in formulae: treat a compound ion as a single unit. If more than one is needed, put it in **brackets** with the number outside, because the formula of the ion itself cannot be changed. Brackets are not needed if there is only one compound ion.

Worked Example C3.03:

- Sodium carbonate: Na^+ (1+) and CO_3^{2-} (2−). Cross over: **Na_2CO_3** (only one carbonate ion, so no brackets).
- Ammonium sulfate: NH_4^+ (1+) and SO_4^{2-} (2−). Cross over: **$(NH_4)_2SO_4$**.

Compound	Formula	Ions present	Ratio
Sodium chloride	NaCl	Na^+, Cl^-	1 : 1
Ammonium nitrate	NH_4NO_3	NH_4^+, NO_3^-	1 : 1
Potassium sulfate	K_2SO_4	K^+, SO_4^{2-}	2 : 1
Calcium hydrogencarbonate	$Ca(HCO_3)_2$	Ca^{2+}, HCO_3^-	1 : 2
Copper(II) sulfate	$CuSO_4$	Cu^{2+}, SO_4^{2-}	1 : 1
Magnesium nitrate	$Mg(NO_3)_2$	Mg^{2+}, NO_3^-	1 : 2
Aluminium chloride	$AlCl_3$	Al^{3+}, Cl^-	1 : 3

Word Equations

Some reactions are spectacular (explosives, or the "volcano" reaction in which ammonium dichromate decomposes, giving out heat and light), while others, such as neutralising an acid with an alkali, show no visible change. Every reaction can be summarised by an equation.

- A **word equation** links the names of the **reactants** (the substances that react) with the names of the **products** (the new substances formed): reactants → products.
- magnesium + oxygen → magnesium oxide

- hydrogen + oxygen → water (a highly exothermic reaction, used in the hydrogen–oxygen fuel cell)
- Energy is **not** included in an equation, even when a lot is released, because energy is not a chemical substance.

Balanced Symbol Equations

- Equations become more useful when they are written with **chemical formulae**.
- The **law of conservation of mass**: the total mass of the products is always equal to the total mass of the reactants.
- During a reaction bonds are broken and new bonds are made, so the atoms are **rearranged**. No atoms are created or lost, and atoms do not change into atoms of other elements.
- So a **balanced equation** has the same number of each type of atom on both sides, e.g. $2H_2 + O_2 \rightarrow 2H_2O$ has 4 H atoms and 2 O atoms on each side. Where no number is written in front of a formula, one is meant.

Rules for balancing:

- Write the correct formula for every reactant and product first.
- **Never change a formula** (the small numbers) to make an equation balance; formulae are fixed by the bonding in the substance.
- Balance only by putting **multiplying numbers in front** of formulae.
- Count every type of atom on each side at the end to check.

Example, potassium with water (alkali metal + water → metal hydroxide + hydrogen):

- potassium + water → potassium hydroxide + hydrogen
- $K + H_2O \rightarrow KOH + H_2$ is not balanced: there are 2 H atoms on the left but 3 on the right.
- On the left, hydrogen occurs only in H_2O, so the hydrogen atoms must come in pairs. Double the KOH, then double the K and the H_2O on the left.
- **$2K + 2H_2O \rightarrow 2KOH + H_2$** (2 K, 4 H and 2 O atoms on each side).

Worked Example C3.04 (balancing in four steps, magnesium burning in air):

Step 1: Identify the reactants and products: magnesium burns in air (oxygen) to form magnesium oxide.

Step 2: Write the word equation: magnesium + oxygen → magnesium oxide

Step 3: Write the formulae: $Mg + O_2 \rightarrow MgO$. This is not balanced because oxygen is diatomic: there are two O atoms on the left but only one on the right.

Step 4: Balance the equation: **$2Mg + O_2 \rightarrow 2MgO$**

State Symbols

State symbol	Meaning
(s)	solid
(l)	liquid
(g)	gas
(aq)	aqueous solution (dissolved in water)

- Water produced in a reaction is a liquid, (l), not (aq).
- $Mg(s) + 2HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2(g)$
- $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
- $CuSO_4(aq) + 2NaOH(aq) \rightarrow Cu(OH)_2(s) + Na_2SO_4(aq)$, where the (s) shows that a solid **precipitate** forms.

Ionic Equations

- When ionic substances react in solution, some of the ions do not change; these are called **spectator ions**.
- An **ionic equation** shows only the ions (and other substances) that change during the reaction.

How to write an ionic equation:

- Write the balanced symbol equation with state symbols.
- Split each ionic compound that is (aq) into its separate ions. Do not split solids, water, gases or covalent molecules.
- Cross out the spectator ions, which appear unchanged on both sides.
- Write what is left, and check that the atoms and the charges balance.

Neutralisation (acid + alkali):

$[H^+(aq) + Cl^-(aq)] + [Na^+(aq) + OH^-(aq)] \rightarrow [Na^+(aq) + Cl^-(aq)] + H_2O(l)$

Na^+ and Cl^- are spectator ions, leaving **$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$**. This is the ionic equation for all neutralisation reactions between acids and alkalis.

Precipitation (Figure C3.13):

$Cu^{2+}(aq) + SO_4^{2-}(aq) + 2Na^+(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s) + 2Na^+(aq) + SO_4^{2-}(aq)$

Na^+ and SO_4^{2-} are spectator ions, leaving **$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$**.

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2. Relative Masses of Atoms and Molecules (C3.02)

Relative Atomic Mass (Ar)

- Atoms are extremely light: one hydrogen atom has a mass of 1.7×10^{-24} g (0.000 000 000 000 000 000 001 7 g). Masses in grams are inconvenient, so atoms are compared on a **relative** scale.
- The standard is an atom of the **carbon-12** isotope, which is given a mass of exactly 12 units. On this scale a hydrogen atom has a mass of 1, so one carbon atom has the same mass as 12 hydrogen atoms, and one helium atom (Ar = 4) has the same mass as 4 hydrogen atoms.
- **Relative atomic mass (Ar)** is the **average** mass of an atom of an element, taking into account its naturally occurring **isotopes** (atoms of the same element with different numbers of neutrons), on the scale where carbon-12 = 12.
- Most elements are mixtures of isotopes, so most Ar values are not whole numbers. Values are usually rounded to whole numbers, except **chlorine, Ar = 35.5** (chlorine has isotopes of mass 35 and 37). Fluorine is unusual in having just one isotope, fluorine-19.
- An ion has the same relative mass as its atom; the mass of the electrons lost or gained is so small that it is ignored.

Ar values used in this guide:

Element	Ar	Element	Ar
Hydrogen, H	1	Sulfur, S	32
Carbon, C	12	Chlorine, Cl	35.5
Nitrogen, N	14	Potassium, K	39
Oxygen, O	16	Calcium, Ca	40
Sodium, Na	23	Iron, Fe	56
Magnesium, Mg	24	Cobalt, Co	59
Aluminium, Al	27	Copper, Cu	64
Silicon, Si	28	Bromine, Br	80

Relative Molecular Mass and Relative Formula Mass (Mr)

- **Relative molecular mass (Mr)**: the sum of the Ar values of all the atoms in one **molecule**. It is used for covalent substances made of molecules.
- **Relative formula mass (Mr)**: the sum of the Ar values of all the atoms in one **formula unit**. It is used for ionic compounds, which are giant structures with no molecules.
- Ar and Mr are relative values, so they have **no units**.
- Examples: Mr(H_2) = 2 × 1 = 2; Mr(H_2O) = (2 × 1) + 16 = 18; Mr(NaCl) = 23 + 35.5 = 58.5 (formula unit Na^+Cl^-).

Substance	Formula	Atoms in the formula	Mr
Hydrogen	H_2	2 H	2 × 1 = 2
Carbon dioxide	CO_2	1 C, 2 O	12 + (2 × 16) = 44
Calcium carbonate	$CaCO_3$	1 Ca, 1 C, 3 O	40 + 12 + (3 × 16) = 100
Ammonium sulfate	$(NH_4)_2SO_4$	2 N, 8 H, 1 S, 4 O	(2 × 14) + (8 × 1) + 32 + (4 × 16) = 132
Hydrated magnesium sulfate	$MgSO_4 \cdot 7H_2O$	1 Mg, 1 S, 4 O, plus 7 H_2O (14 H, 7 O)	24 + 32 + (4 × 16) + (14 × 1) + (7 × 16) = 246

- A number after a bracket multiplies everything inside it: $(NH_4)_2$ contains 2 N and 8 H atoms.
- In a hydrated salt, the number in front of H_2O multiplies the whole water molecule: $7H_2O$ contains 14 H and 7 O atoms.

Compound Formation and Chemical Formulae

- Samples of the same compound, however they are made, always contain the **same elements** combined in the **same ratio by mass**.

Experiment: making magnesium oxide (Figure C3.16)

- Weigh an empty crucible with its lid, then reweigh it with a coil of magnesium ribbon inside.
- Heat the crucible strongly. Lift the lid very carefully from time to time to let air in, but do not let any solid escape as white smoke.
- Let the crucible cool, then reweigh it. The **increase in mass** is the mass of oxygen that has combined with the magnesium.

Worked Example C3.05 (typical results):

Measurement	Mass (g)
(a) Crucible + lid	8.52
(b) Crucible + lid + magnesium	8.88
(c) Crucible + lid + magnesium oxide	9.12
Mass of magnesium = (b) − (a)	0.36
Mass of magnesium oxide = (c) − (a)	0.60
Mass of oxygen = 0.60 − 0.36	0.24

- When the class results are plotted as mass of oxygen (y-axis) against mass of magnesium (x-axis), they give a **straight line through the origin** (Figure C3.17). The masses are always in the same ratio, so a definite compound with a fixed formula is formed.
- Magnesium oxide always contains 60% magnesium and 40% oxygen by mass. Ammonium nitrate always contains 35% nitrogen, 60% oxygen and 5% hydrogen by mass.

Calculating Reacting Masses

Because atoms are conserved, the balanced equation and the Ar values give the masses that react:

$2Mg + O_2 \rightarrow 2MgO$

$(2 \times 24) + (2 \times 16) \rightarrow (2 \times 24) + (2 \times 16)$

$48\ g + 32\ g \rightarrow 80\ g$

So 48 g of magnesium reacts with 32 g of oxygen to give 80 g of magnesium oxide. Any masses in the same ratio react in the same way.

Worked Example C3.06 (simple proportion): 0.24 g of magnesium reacts with 0.16 g of oxygen to produce 0.40 g of magnesium oxide. How much magnesium oxide is produced by burning 12 g of magnesium?

- 0.24 g of Mg produces 0.40 g of MgO.
- So 1 g of Mg produces 0.40 ÷ 0.24 = 1.67 g of MgO.
- So 12 g of Mg produces 12 × 1.67 = **20 g of MgO**.

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3. The Mole and the Avogadro Constant (C3.03)

Why Chemists Use the Mole

- Atoms and molecules are far too small to weigh out or count one at a time, so chemists need a "counting unit". Banks do the same with coins: they weigh them rather than count them, because a fixed number of a particular coin always has the same mass.
- Chemists count particles by weighing them. For example, **concentration** (the amount of a substance in a given volume) can be measured in **g/dm³**.
- A mass in grams does not directly tell us the number of particles, because atoms of different elements have different masses. So chemists measure amounts in **moles**.

The Mole, Molar Mass and the Avogadro Constant

- One **mole** of a substance has a mass equal to its **Ar or Mr in grams**, and contains **6.02×10^{23}** particles: atoms, molecules or formula units, depending on the substance.
- The mass of one mole is the **molar mass**, measured in **g/mol**.
- 6.02×10^{23} is the **Avogadro constant** (symbol L), named after the 19th-century Italian chemist Amedeo Avogadro.
- Example: ethanol has Mr = 46, so 46 g of ethanol is 1 mole and contains 6.02×10^{23} ethanol molecules.

Substance	Formula	Ar or Mr	Mass of 1 mole	Particles in 1 mole
Carbon	C	12	12 g	6.02×10^{23} C atoms
Iron	Fe	56	56 g	6.02×10^{23} Fe atoms
Hydrogen	H_2	2 × 1 = 2	2 g	6.02×10^{23} H_2 molecules
Oxygen	O_2	2 × 16 = 32	32 g	6.02×10^{23} O_2 molecules
Water	H_2O	(2 × 1) + 16 = 18	18 g	6.02×10^{23} H_2O molecules
Magnesium oxide	MgO	24 + 16 = 40	40 g	6.02×10^{23} MgO formula units
Calcium carbonate	$CaCO_3$	40 + 12 + (3 × 16) = 100	100 g	6.02×10^{23} $CaCO_3$ formula units

Substance	Formula	Ar or Mr	Mass of 1 mole	Particles in 1 mole
Silicon(IV) oxide	SiO_2	28 + (2 × 16) = 60	60 g	6.02×10^{23} SiO_2 formula units

Mass Calculations

Finding the molar mass of a substance:

Step 1: Write down the formula, e.g. ethanol is C_2H_5OH.

Step 2: Work out Mr. Ethanol contains 2 C atoms (Ar = 12), 6 H atoms (Ar = 1) and 1 O atom (Ar = 16), so Mr = (2 × 12) + (6 × 1) + 16 = 46.

Step 3: Express this in grams per mole: the molar mass of ethanol is 46 g/mol.

The key equation:

amount of substance (mol) = mass (g) ÷ molar mass (g/mol)

The calculation triangle (Figure C3.20): mass is written at the top, with number of moles and molar mass side by side at the bottom (multiplied together). Cover up the quantity you want:

- number of moles = mass ÷ molar mass
- mass = number of moles × molar mass
- molar mass = mass ÷ number of moles

Worked Example C3.07: How many moles are there in 60 g of sodium hydroxide, NaOH?

- Mr(NaOH) = 23 + 16 + 1 = 40, so the molar mass is 40 g/mol.
- number of moles = mass ÷ molar mass = 60 g ÷ 40 g/mol = **1.5 mol**

Calculating Reacting Amounts

- A balanced equation tells us not only what the products are, but also how much product to expect from given amounts of reactants.
- Fe + S → FeS: 1 mol of iron (56 g) reacts with 1 mol of sulfur (32 g) to give 1 mol of iron(II) sulfide (88 g). The total mass is unchanged (56 + 32 = 88), as the law of conservation of mass requires.
- The masses of iron and sulfur must always be in the ratio 56 : 32, e.g. 5.6 g Fe + 3.2 g S → 8.8 g FeS.
- If 5.0 g of sulfur is reacted with 5.6 g of iron, only 3.2 g of the sulfur can react, so 5.0 − 3.2 = 1.8 g of sulfur remains unreacted.

The chemical "bridge" (Figure C3.21), following "up, across, down":

- **Up:** change the mass of the substance you know into moles (moles = mass ÷ molar mass).
- **Across:** use the mole ratio from the balanced equation to find the moles of the substance you want.
- **Down:** change those moles into a mass (mass = moles × molar mass).
- The bridge works in either direction: from reactant to product, or from product back to reactant.

Worked Example C3.08: What mass of aluminium oxide is produced when 9.2 g of aluminium reacts completely with oxygen gas?

Balanced equation: $4Al + 3O_2 \rightarrow 2Al_2O_3$, so the ratio Al : Al_2O_3 = 4 mol : 2 mol.

Step 1 (up): number of moles of Al = 9.2 g ÷ 27 g/mol = 0.34 mol

Step 2 (across): 4 mol of Al produce 2 mol of Al_2O_3, so 0.34 mol of Al produce 0.17 mol of Al_2O_3.

Step 3 (down): Mr(Al_2O_3) = (2 × 27) + (3 × 16) = 102, so mass of Al_2O_3 = 0.17 mol × 102 g/mol = **17.3 g**

- Read reacting-mass questions carefully and set out the calculation using the balanced equation, so that you can see which substances matter. Always use the balancing numbers: this is the **stoichiometry** of the equation.

- Keeping the unrounded number of moles (9.2 ÷ 27 = 0.3407… mol) in your calculator gives 17.4 g, so round only at the end.

Limiting Reactants

- Reactants are not always mixed in the exact amounts shown by the equation. Sometimes too much of one reactant (an **excess**) is used on purpose.
- The **limiting reactant** is the reactant that is **not in excess**. It is used up completely, the reaction stops when it runs out, and it decides how much product is formed. The other reactant is left over and is said to be **in excess**.
- Figure C3.22 shows 2A + B → A_2B: when all of A has reacted, some B is left over, so A is the limiting reactant.
- In Worked Example C3.08 an excess of oxygen made sure that all the aluminium reacted, so aluminium was the limiting reactant.

How to find the limiting reactant:

- Work out the number of moles of each reactant.
- Divide each by its balancing number (coefficient) in the balanced symbol equation.
- The **smallest** number shows the limiting reactant.

Worked Example C3.09: 6.0 g of cobalt(II) carbonate was reacted with 0.08 mol of hydrochloric acid to produce cobalt(II) chloride. Show that the cobalt(II) carbonate was in excess.

$CoCO_3(s) + 2HCl(aq) \rightarrow CoCl_2(aq) + H_2O(l) + CO_2(g)$ (Ar: Co = 59, C = 12, O = 16)

- $Mr(CoCO_3) = 59 + 12 + (3 \times 16) = 119$, so the molar mass is 119 g/mol.
- number of moles of $CoCO_3$ = 6.0 g ÷ 119 g/mol = 0.05 mol
- 1 mol of $CoCO_3$ reacts with 2 mol of HCl, so 0.05 mol of $CoCO_3$ would react with 0.10 mol of HCl.
- Only 0.08 mol of HCl was present, so the acid is used up first and the cobalt(II) carbonate is **in excess**.
- Coefficient check: $CoCO_3$ gives 0.05 ÷ 1 = 0.05 and HCl gives 0.08 ÷ 2 = 0.04. HCl gives the smaller number, so it is the limiting reactant.

Mole Calculations Involving Gases

- Weighing a gas is difficult, but measuring its volume is easy.
- **Avogadro's law:** equal volumes of any gases, at the same temperature and pressure, contain the same number of particles.
- So one mole of **any** gas occupies the same volume: about **24 dm³ (24 litres)** at **room temperature and pressure (r.t.p.)**, which is 25 °C (298 K) and 101.3 kPa (1 atmosphere). This is the **molar gas volume**.
- 1 dm³ = 1 litre = 1000 cm³, so the molar gas volume is also 24 000 cm³.

Gas	Formula	Molar mass (g/mol)	Molar gas volume at r.t.p. (dm³/mol)	Particles in 1 mole
Hydrogen	H_2	2	24	6.02×10^{23} molecules
Oxygen	O_2	32	24	6.02×10^{23} molecules
Carbon dioxide	CO_2	44	24	6.02×10^{23} molecules
Ethane	C_2H_6	30	24	6.02×10^{23} molecules

The key equation:

number of moles of gas = volume (dm^3) ÷ molar gas volume (24 dm^3/mol at r.t.p.)

Worked Example C3.10: If 8 g of sulfur is burnt, what volume of SO_2 is produced?

$S(s) + O_2(g) \rightarrow SO_2(g)$: 1 mol of S (32 g) reacts with 1 mol of O_2 (24 dm^3) to give 1 mol of SO_2 (24 dm^3).

- number of moles of sulfur = 8 g ÷ 32 g/mol = 0.25 mol
- From the equation, 1 mol of S gives 1 mol of SO_2, so 0.25 mol of S gives 0.25 mol of SO_2.
- volume of SO_2 = 0.25 mol × 24 dm^3/mol = **6 dm^3 at r.t.p.**

Reactions between gases: gas volumes react and form in the **same ratio as the numbers of moles** in the equation.

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

1 mol (24 dm^3) + 1 mol (24 dm^3) → 2 mol (48 dm^3), which is 1 volume : 1 volume : 2 volumes.

So 20 cm^3 of hydrogen reacting with enough chlorine produces 40 cm^3 of hydrogen chloride gas.

Concentration in g/dm^3

- **Concentration** is the amount of a substance (the solute) dissolved in a given volume of solution.
- concentration (g/dm^3) = mass of solute (g) ÷ volume of solution (dm^3)
- Convert cm^3 into dm^3 first by dividing by 1000. For example, for 5 g dissolved in 250 cm^3 of solution: volume = 250 ÷ 1000 = 0.250 dm^3, so concentration = 5 g ÷ 0.250 dm^3 = 20 g/dm^3.

Green Chemistry

- The mole and the ratios in which chemicals react are used to decide the conditions for key industrial processes, such as the Haber and Contact processes. These ideas are part of the **Green Chemistry** approach, set out in 12 principles.
- The principles include: maximising the amount of the raw materials that ends up in the desired product (measured as **atom economy**); using renewable raw materials and energy sources; using the least environmentally harmful substances and solvents; using energy-efficient processes; and avoiding the production of waste.

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4. Key Terms

Term	Definition
Stoichiometry	Working out the amounts of substances that react and are produced in chemical reactions, using formulae, balanced equations and moles ("chemical accountancy")
Chemical symbol	A letter or group of letters representing an element in a chemical formula
Chemical formula	A combination of symbols and numbers showing the elements in a substance and the number of atoms (or the ratio of ions) of each element present
Diatomic molecule	A molecule made of two atoms bonded together, e.g. H_2, N_2, O_2, Cl_2
Compound ion	An ion made up of several different atoms covalently bonded together and with an overall charge; also called a molecular ion (negatively charged compound ions containing oxygen, such as SO_4^{2-}, are called oxyanions)
Valency	The combining power of an atom: the number of bonds it forms (Groups I–IV = group number; Groups V–VII = 8 − group number)

Term	Definition
Molecular formula	A formula that shows the actual number of atoms of each element present in a molecule of the compound
Formula unit	The molecule or group of ions defined by the chemical formula of the substance, e.g. one Na^+ ion and one Cl^- ion for NaCl
Word equation	A summary of a chemical reaction using the chemical names of the reactants and products
Reactants	The chemical substances that react together in a chemical reaction
Products	The substance(s) produced by a chemical reaction
Law of conservation of mass	The total mass of the products of a reaction is always equal to the total mass of the reactants, because atoms are rearranged but not created or destroyed
Balanced chemical equation	A summary of a chemical reaction using chemical formulae; the total number of any of the atoms involved is the same on both the reactant and product sides of the equation
State symbols	Symbols used to show the physical state of the reactants and products in a chemical reaction: (s) solid, (l) liquid, (g) gas, (aq) aqueous solution
Aqueous solution	A solution in which the solvent is water, shown by the state symbol (aq)
Ionic equation	The simplified equation for a reaction involving ionic substances: only those ions which change during the reaction are shown
Spectator ions	Ions that are present in the reaction mixture but do not change during the reaction, so they are left out of the ionic equation
Precipitate	An insoluble solid formed when two solutions are mixed, e.g. $Cu(OH)_2(s)$
Isotopes	Atoms of the same element that have the same number of protons but different numbers of neutrons
Relative atomic mass (Ar)	The average mass of naturally occurring atoms of an element on a scale where an atom of carbon-12 has a mass of exactly 12 units
Relative molecular mass (Mr)	The sum of all the relative atomic masses of the atoms present in a molecule
Relative formula mass (Mr)	The sum of all the relative atomic masses of the atoms present in a formula unit of a substance
Mole	The measure of amount of substance in chemistry; 1 mole of a substance has a mass equal to its relative formula mass in grams, and contains 6.02×10^{23} (the Avogadro constant) atoms, molecules or formula units, depending on the substance
Avogadro constant	The number (6.02×10^{23}) of characteristic particles in 1 mole of a substance; symbol L
Molar mass	The mass, in grams, of 1 mole of a substance; unit g/mol
Limiting reactant	The reactant that is not in excess; it is completely used up and so decides the amount of product formed
Excess	More of a reactant than is needed to react with the other reactant(s), so some is left over at the end of the reaction

Term	Definition
Avogadro's law	Equal volumes of any gas, under the same conditions of temperature and pressure, contain the same number of particles
Room temperature and pressure (r.t.p.)	The standard values of 25 °C (298 K) and 101.3 kPa (1 atmosphere) pressure
Molar gas volume	The volume of 1 mole of any gas under specific conditions of temperature and pressure (24 dm^3 at r.t.p.)
Concentration	The amount of a substance dissolved in a given volume of solution, e.g. measured in g/dm^3
Green Chemistry	An approach to industrial chemistry, set out in 12 principles, that reduces waste and the use or production of hazardous substances
Atom economy	A measure of how much of the raw materials ends up in the desired product

5. Key Equations

Relative Masses

1. Mr = sum of the Ar values of all the atoms in the formula (no units)
2. Percentage of an element by mass = (number of atoms of the element × Ar) ÷ Mr × 100
3. Reacting masses by proportion: mass of product = mass of reactant used × (known mass of product ÷ known mass of reactant)

The Mole

1. Amount of substance (mol) = mass (g) ÷ molar mass (g/mol)
2. Mass (g) = amount (mol) × molar mass (g/mol)
3. Molar mass (g/mol) = mass (g) ÷ amount (mol), and it is numerically equal to Ar or Mr
4. Avogadro constant, L = 6.02×10^{23} particles per mole
5. Number of particles = amount (mol) × 6.02×10^{23}

Reacting Amounts

1. The bridge: mass of A ÷ molar mass of A = moles of A; moles of A × (ratio B : A from the equation) = moles of B; moles of B × molar mass of B = mass of B
2. Limiting reactant: divide the moles of each reactant by its balancing number; the smallest value is the limiting reactant

Gases at r.t.p. (25 °C and 101.3 kPa)

1. Amount of gas (mol) = volume (dm^3) ÷ 24 dm^3/mol
2. Volume of gas (dm^3) = amount (mol) × 24 dm^3/mol
3. Amount of gas (mol) = volume (cm^3) ÷ 24 000 cm^3/mol
4. 1 dm^3 = 1 litre = 1000 cm^3
5. For reacting gases, the ratio of volumes = the ratio of moles in the balanced equation

Concentration

1. Concentration (g/dm^3) = mass of solute (g) ÷ volume of solution (dm^3)
2. Volume (dm^3) = volume (cm^3) ÷ 1000

Key Balanced Equations

1. $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
2. $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
3. $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$
4. $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
5. $Mg(s) + 2HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2(g)$
6. $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
7. $CuSO_4(aq) + 2NaOH(aq) \rightarrow Cu(OH)_2(s) + Na_2SO_4(aq)$
8. $Fe(s) + S(s) \rightarrow FeS(s)$
9. $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$
10. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
11. $CoCO_3(s) + 2HCl(aq) \rightarrow CoCl_2(aq) + H_2O(l) + CO_2(g)$
12. $S(s) + O_2(g) \rightarrow SO_2(g)$
13. $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$
14. $4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$
15. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

Ionic Equations

1. $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ (every acid–alkali neutralisation)
2. $Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$
3. $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
4. $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

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6. Practice Questions with Full Answers

Before You Start — Question 1

Knowledge of the symbols of the elements and their use in chemical formulae is important in building confidence in chemistry. Think about the symbols of elements and any formulae of compounds you have met when learning science, and write a list of as many as you can think of.

Answer:

A good list includes both elements and compounds, for example:

- Elements: hydrogen H, helium He, carbon C, nitrogen N, oxygen O, sodium Na, magnesium Mg, aluminium Al, sulfur S, chlorine Cl, potassium K, calcium Ca, iron Fe, copper Cu, zinc Zn, silver Ag, gold Au, lead Pb.
- Elements that exist as molecules: hydrogen H_2, nitrogen N_2, oxygen O_2, chlorine Cl_2.
- Compounds: water H_2O, carbon dioxide CO_2, methane CH_4, ammonia NH_3, sodium chloride NaCl, hydrochloric acid HCl, sulfuric acid H_2SO_4, sodium hydroxide NaOH, calcium carbonate $CaCO_3$, magnesium oxide MgO, copper(II) sulfate $CuSO_4$, glucose $C_6H_{12}O_6$.

Before You Start — Question 2

Based on this list, create five questions to ask a partner.

Answer:

Five example questions, with their answers:

- What is the symbol for potassium, and why is it not P? (K, from its Latin name kalium; P is the symbol for phosphorus.)
- Which elements, and how many atoms of each, are in one molecule of sulfuric acid, H_2SO_4? (2 hydrogen atoms, 1 sulfur atom and 4 oxygen atoms.)
- Name the compound with the formula $CaCO_3$. (Calcium carbonate.)

- Why is oxygen gas written as O_2 and not O? (Oxygen exists as diatomic molecules, each made of two atoms bonded together.)
- What is the difference between Co and CO? (Co is the element cobalt; CO is carbon monoxide, a compound of carbon and oxygen.)

Before You Start — Question 3

Working in pairs, take it in turns to ask each other your questions. How did you do? Discuss with your partner what you know about the elements and formulae included in your questions.

Answer:

Model answer: We both knew the common symbols (H, O, C, N, Cl) and simple formulae such as H_2O and CO_2. The symbols that come from Latin names (Na, K, Fe, Cu, Ag, Au, Pb) were harder to remember. We agreed on the key rules: a symbol starts with a capital letter and any second letter is lower case; the small subscript numbers in a formula show how many atoms of the element just before them are present; and gases such as hydrogen, nitrogen, oxygen and chlorine are written as diatomic molecules (H_2, N_2, O_2, Cl_2).

Science in Context C3.01 — Discussion Question 1

Stoichiometry is "chemical accountancy": it lets industrial chemists calculate how much of each reactant is needed and how much product will be made. The 12 principles of Green Chemistry aim to reduce the use or production of hazardous or waste substances, and modern chemical plants are designed for safety and atom economy. What costs, other than those of the raw materials, should be considered when considering the economy of an industrial process?

Answer:

- Energy: fuel or electricity for heating, for producing high pressures, and for pumping and cooling.
- Building, running and maintaining the plant and its equipment, including replacing catalysts.
- Labour: the wages of the workers who run the plant.
- Transporting the raw materials to the plant and the products to customers.
- Water, for example for cooling.
- Separating and purifying the product.
- Treating and safely disposing of waste and unwanted by-products.
- Safety measures, pollution control and meeting environmental regulations.

Science in Context C3.01 — Discussion Question 2

In what way could the economic problem of by-products be reduced by positive and imaginative marketing?

Answer: Instead of paying to dispose of by-products as waste, a company can find uses for them and market them as useful products, for example by selling them to other industries as raw materials or turning them into products such as fertilisers, fuels or building materials. Advertising that the process wastes very little can also improve the company's "green" image. The by-products then earn money instead of costing money, so the whole process becomes more economical and less wasteful.

Activity C3.01 — Question 1

Jigsaw compounds: work in groups to make a set of jigsaw cards (using paper, scissors and pens) for the ions in the ionic compounds of Table C3.03 (see Section 1), plus extra cards for CO_3^{2-}, O^{2-} and Fe^{3+} ions. Positive-ion cards have "slots" and negative-ion cards have "tabs" that fit into the slots; the number of slots or tabs on a card equals the charge on the ion. The cards for a compound fit together to make a complete rectangle, and more than one card of the same ion may be needed. Having made your set of cards, use them to work out the formulae of the following compounds: copper carbonate, iron(III) oxide and ammonium sulfate.

Answer:

- **Copper carbonate:** a Cu^{2+} card has 2 slots and a CO_3^{2-} card has 2 tabs, so one of each fits together. Formula: **$CuCO_3$**.
- **Iron(III) oxide:** an Fe^{3+} card has 3 slots and an O^{2-} card has 2 tabs. The smallest number of slots and tabs that match is 6, so 2 Fe^{3+} cards (6 slots) fit with 3 O^{2-} cards (6 tabs). Formula: **Fe_2O_3**.
- **Ammonium sulfate:** an NH_4^+ card has 1 slot and an SO_4^{2-} card has 2 tabs, so 2 NH_4^+ cards fit with 1 SO_4^{2-} card. Formula: **$(NH_4)_2SO_4$**, with brackets because there are two ammonium ions.

Activity C3.01 — Question 2

Then challenge each other in the group to make a compound you name.

Answer:

Examples of compounds that can be made from the cards:

Compound named	Cards needed	Formula
Calcium oxide	1 Ca^{2+} + 1 O^{2-}	CaO
Sodium oxide	2 Na^+ + 1 O^{2-}	Na_2O
Sodium carbonate	2 Na^+ + 1 CO_3^{2-}	Na_2CO_3
Iron(III) chloride	1 Fe^{3+} + 3 Cl^-	$FeCl_3$
Aluminium oxide	2 Al^{3+} + 3 O^{2-}	Al_2O_3
Magnesium nitrate	1 Mg^{2+} + 2 NO_3^-	$Mg(NO_3)_2$
Iron(III) sulfate	2 Fe^{3+} + 3 SO_4^{2-}	$Fe_2(SO_4)_3$

Activity C3.01 — Question 3

Using some of your examples, compare this method to the cross-over method described earlier in this chapter.

Answer:

- Both methods give the same formulae. For iron(III) oxide, crossing over the charges of Fe^{3+} and O^{2-} gives Fe_2O_3, the same as fitting 2 Fe^{3+} cards with 3 O^{2-} cards.
- The jigsaw method is visual and hands-on: every slot must be filled by a tab, which shows why the total positive and negative charges must balance. It also shows that a compound ion such as NH_4^+ or SO_4^{2-} is a single unit that cannot be split, which is why brackets are needed in $(NH_4)_2SO_4$.
- The cross-over method is quicker and needs no equipment, so it is the method to use in an exam. However, it can give a ratio that must be simplified (Ca^{2+} and O^{2-} cross over to Ca_2O_2, which simplifies to CaO), whereas the jigsaw gives the simplest ratio directly (one Ca^{2+} card fits one O^{2-} card).

Activity C3.01 — Self-Assessment

Think about the following questions and give yourself a happy, neutral or sad face. How confident did you feel working out and writing the formulae of ionic compounds; working out the formula of copper carbonate; working out the formula of iron(III) oxide; working out the formula of ammonium sulfate? Do you find this method of finding the formula of an ionic compound more helpful than the cross-over method?

Answer:

- Check your formulae against the correct answers: $CuCO_3$, Fe_2O_3 and $(NH_4)_2SO_4$. If all three were right, you can feel confident; if not, revise the charges on the ions and how brackets are used for compound ions.

- Many students find the jigsaw method more helpful for understanding why the charges must balance, and the cross-over method faster once the idea is understood. A good strategy is to use the cross-over method in exams and then check that the total positive charge equals the total negative charge (for Fe_2O_3: 2 × 3+ = 6+ and 3 × 2− = 6−).

Worked Example C3.04 — Question

Worked Example C3.04 balances an equation in four steps: identify the reactants and products; write the word equation; write the formulae; balance the equation. Now work through the steps to find the balanced symbol equation for the reaction that takes place when burning sodium is lowered into a gas jar of chlorine gas. Sodium chloride is produced in this reaction. Remember that chlorine is a diatomic gas.

Answer:

Step 1: The reactants are sodium and chlorine; the product is sodium chloride.

Step 2: sodium + chlorine → sodium chloride

Step 3: $Na + Cl_2 \rightarrow NaCl$. This is not balanced: chlorine is diatomic, so there are 2 Cl atoms on the left but only 1 on the right.

Step 4: Put a 2 in front of NaCl to balance the chlorine ($Na + Cl_2 \rightarrow 2NaCl$), then put a 2 in front of Na to balance the sodium:

$2Na + Cl_2 \rightarrow 2NaCl$

With state symbols: $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$. Check: there are 2 Na atoms and 2 Cl atoms on each side.

Question C3.01

Write word equations for the reactions described in (a)–(c).

(a) Iron rusts because it reacts with oxygen in the air to form a compound called iron(III) oxide.

(b) Sodium hydroxide neutralises sulfuric acid to form sodium sulfate and water.

(c) Sodium reacts strongly with water to give a solution of sodium hydroxide; hydrogen gas is also given off.

Answer:

(a) iron + oxygen → iron(III) oxide

(b) sodium hydroxide + sulfuric acid → sodium sulfate + water

(c) sodium + water → sodium hydroxide + hydrogen

Question C3.02

Balance the following symbol equations by filling in the missing numbers:

(a) $...Cu + O_2 \rightarrow ...CuO$

(b) $N_2 + ...H_2 \rightarrow ...NH_3$

(c) $...Na + O_2 \rightarrow ...Na_2O$

(d) $...NaOH + H_2SO_4 \rightarrow Na_2SO_4 + ...H_2O$

(e) $...Al + ...Cl_2 \rightarrow ...AlCl_3$

(f) $...Fe + ...H_2O \rightarrow ...Fe_3O_4 + ...H_2$

Answer:

(a) $2Cu + O_2 \rightarrow 2CuO$ (2 Cu and 2 O atoms on each side)

(b) $N_2 + 3H_2 \rightarrow 2NH_3$ (2 N and 6 H atoms on each side)

(c) $4Na + O_2 \rightarrow 2Na_2O$ (4 Na and 2 O atoms on each side)

(d) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$ (2 Na, 1 S, 6 O and 4 H atoms on each side)

(e) $2Al + 3Cl_2 \rightarrow 2AlCl_3$ (2 Al and 6 Cl atoms on each side)

(f) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ (3 Fe, 8 H and 4 O atoms on each side; the number in front of Fe_3O_4 is 1, so it is not written)

Question C3.03

Use these word equations to write balanced symbol equations.

(a) hydrogen + chlorine → hydrogen chloride

(b) copper + oxygen → copper oxide

(c) magnesium + zinc chloride → magnesium chloride + zinc

Answer:

(a) $H_2 + Cl_2 \rightarrow 2HCl$ (hydrogen and chlorine are both diatomic; 2 H and 2 Cl atoms on each side)

(b) $2Cu + O_2 \rightarrow 2CuO$ (the copper oxide is copper(II) oxide, made of Cu^{2+} and O^{2-} ions; 2 Cu and 2 O atoms on each side)

(c) $Mg + ZnCl_2 \rightarrow MgCl_2 + Zn$ (Mg^{2+} and Zn^{2+} each combine with two Cl^- ions; 1 Mg, 1 Zn and 2 Cl atoms on each side)

Question C3.04

Explain the meaning of the symbols (s), (l), (aq) and (g) in the following equation, with reference to each reactant and product:

$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

Answer:

- $Na_2CO_3(s)$: the sodium carbonate is a **solid**.
- $HCl(aq)$: the hydrochloric acid is **aqueous** (hydrogen chloride dissolved in water).
- $NaCl(aq)$: the sodium chloride produced is **aqueous** (it is dissolved in the water, forming a solution).
- $H_2O(l)$: the water produced is a **liquid**.
- $CO_2(g)$: the carbon dioxide produced is a **gas**, given off as bubbles.

Question C3.05

Write an ionic equation, including state symbols, for each of the following reactions. (Note: you will have to initially construct a balanced symbol equation.)

(a) silver nitrate solution + sodium chloride solution → silver chloride + sodium nitrate solution

(b) sodium sulfate solution + barium nitrate solution → sodium nitrate solution + barium sulfate

Answer:

(a) Balanced symbol equation: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

As ions: $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

Na^+ and NO_3^- are spectator ions, so the ionic equation is: $\mathbf{Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)}$

(b) Balanced symbol equation: $Na_2SO_4(aq) + Ba(NO_3)_2(aq) \rightarrow 2NaNO_3(aq) + BaSO_4(s)$

As ions: $2Na^+(aq) + SO_4^{2-}(aq) + Ba^{2+}(aq) + 2NO_3^-(aq) \rightarrow 2Na^+(aq) + 2NO_3^-(aq) + BaSO_4(s)$

Na^+ and NO_3^- are spectator ions, so the ionic equation is: $\mathbf{Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)}$

In both reactions the product that is not a solution (silver chloride, barium sulfate) is an insoluble solid precipitate, so it has the state symbol (s) and is not split into ions.

Question C3.06

Write full symbol equations for the following reactions, followed by the ionic equation for the reaction (include state symbols in both types of equation).

(a) dilute hydrochloric acid + potassium hydroxide solution → potassium chloride solution + water

(b) dilute hydrochloric acid + copper carbonate → copper chloride solution + water + carbon dioxide

Answer:

(a) Symbol equation: $HCl(aq) + KOH(aq) \rightarrow KCl(aq) + H_2O(l)$

As ions: $H^+(aq) + Cl^-(aq) + K^+(aq) + OH^-(aq) \rightarrow K^+(aq) + Cl^-(aq) + H_2O(l)$

K^+ and Cl^- are spectator ions, so the ionic equation is: $\mathbf{H^+(aq) + OH^-(aq) \rightarrow H_2O(l)}$

(b) Symbol equation: $2HCl(aq) + CuCO_3(s) \rightarrow CuCl_2(aq) + H_2O(l) + CO_2(g)$

Copper carbonate is an insoluble solid, so it is not split into ions: $2H^+(aq) + 2Cl^-(aq) + CuCO_3(s) \rightarrow Cu^{2+}(aq) + 2Cl^-(aq) + H_2O(l) + CO_2(g)$

Cl^- is the only spectator ion, so the ionic equation is: $\mathbf{2H^+(aq) + CuCO_3(s) \rightarrow Cu^{2+}(aq) + H_2O(l) + CO_2(g)}$

Check: the atoms balance (2 H, 1 Cu, 1 C and 3 O on each side) and so do the charges (2+ on each side).

Worked Example C3.06 — Question

Using the method based on simple proportions, calculate the mass of calcium oxide formed if 8.0 g of calcium carbonate are decomposed completely by heating. A previous experiment had shown that 20.0 g of calcium carbonate produces 11.2 g of calcium oxide when heated.

Answer:

20.0 g of calcium carbonate produces 11.2 g of calcium oxide.

So 1 g of calcium carbonate produces 11.2 ÷ 20.0 = 0.56 g of calcium oxide.

So 8.0 g of calcium carbonate produces 8.0 × 0.56 = **4.48 g of calcium oxide** (about 4.5 g).

Check using the equation $CaCO_3 \rightarrow CaO + CO_2$ with $Mr(CaCO_3) = 100$ and $Mr(CaO) = 40 + 16 = 56$: 100 g gives 56 g, so 8.0 g gives 8.0 × 56 ÷ 100 = 4.48 g.

Question C3.07

Figure C3.19 represents the structures of six different compounds, A–F:

- A: a carbon atom joined by single bonds to four hydrogen atoms.
- B: a giant lattice of alternating sodium ions (Na^+) and iodide ions (I^-).
- C: a chain of three carbon atoms, $H_2C{=}CH{-}CH_3$: the first carbon (with 2 H atoms) is joined by a double bond to the second carbon (with 1 H atom), which is joined by a single bond to the third carbon (with 3 H atoms).
- D: an iodine atom joined by single bonds to three chlorine atoms.
- E: a bromine atom joined by single bonds to five fluorine atoms.
- F: a hydrogen atom joined by a single bond to a bromine atom, H–Br.

(a) What type of bonding is present in compounds A, C, D, E and F?

(b) What type of bonding is present in compound B?

(c) State the molecular formula or formula unit for each compound A–F.

Answer:

(a) **Covalent bonding.** A, C, D, E and F are simple molecules in which non-metal atoms are joined by covalent bonds (shared pairs of electrons).

(b) **Ionic bonding.** B is a giant ionic lattice of oppositely charged sodium ions (Na^+) and iodide ions (I^-), held together by strong electrostatic attraction.

(c) A: CH_4 (methane); B: NaI (formula unit of sodium iodide, one Na^+ ion for every I^- ion); C: C_3H_6 (propene); D: ICl_3; E: BrF_5; F: HBr.

Question C3.08

Calculate the relative molecular or formula masses (Mr) of the following substances (Ar: H = 1, N = 14, O = 16, S = 32, K = 39, Cu = 64, Br = 80):

(a) oxygen, O_2

(b) ammonia, NH_3

(c) sulfuric acid, H_2SO_4

(d) potassium bromide, KBr

(e) copper nitrate, $Cu(NO_3)_2$

Answer:

(a) $Mr(O_2) = 2 \times 16 =$ **32**

(b) $Mr(NH_3) = 14 + (3 \times 1) =$ **17**

(c) $Mr(H_2SO_4) = (2 \times 1) + 32 + (4 \times 16) = 2 + 32 + 64 =$ **98**

(d) $Mr(KBr) = 39 + 80 =$ **119**

(e) $Mr(Cu(NO_3)_2) = 64 + 2 \times [14 + (3 \times 16)] = 64 + (2 \times 62) = 64 + 124 =$ **188**

(a)–(c) are relative molecular masses, because these substances are covalent molecules; (d) and (e) are relative formula masses, because these substances are ionic.

Question C3.09

Calcium carbonate has the formula $CaCO_3$. What is its relative formula mass (Mr)? (Ar: Ca = 40, C = 12, O = 16)

Answer: $Mr(CaCO_3) = 40 + 12 + (3 \times 16) = 40 + 12 + 48 =$ **100**

Reflection — Balancing Equations and Relative Masses

Do you have a successful strategy to help you balance equations? What aspects do you find easy or difficult? Could you explain how to calculate relative molecular or formula masses to someone else?

Answer:

- A strategy for balancing: write the correct formulae first and never change them; count the atoms of each element on both sides (a small tally table helps); balance the elements that appear in only one substance on each side first, and leave oxygen, hydrogen and any element appearing on its own (such as O_2 or H_2) until last; treat a compound ion such as SO_4 as a single unit if it appears unchanged on both sides; finally, recount every atom.
- Easy and difficult aspects: simple equations such as $2Mg + O_2 \rightarrow 2MgO$ are straightforward. It is harder when an element appears in several substances, as in $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$, or when a diatomic element means that a product has to be doubled and the other side rebalanced.
- Explaining Mr: write the formula; count the atoms of each element (a number after a bracket multiplies everything inside it); multiply the number of atoms of each element by its Ar; add up the results. For example, $Mr(Cu(NO_3)_2) = 64 + 2 \times [14 + (3 \times 16)] = 188$. Mr has no units.

Activity C3.02 — Question 1

Getting to grips with the mole: the mole links the number of particles of a chemical to the relative mass of that chemical; in simple terms, it is the chemist's way of describing the amount of substance present. You are going to prepare a five-minute lesson to explain the mole to your classmates, using the equation for the reaction of lithium with oxygen: $4Li + O_2 \rightarrow 2Li_2O$. Work in a pair to write down an explanation of moles and how they link to mass via Mr. Your explanation should include what the mole is (you might want to include a reference to Avogadro's constant).

Answer: A mole is the chemist's unit for the amount of a substance. One mole of any substance contains 6.02×10^{23} particles (the Avogadro constant), which may be atoms, molecules or formula units, and it has a mass equal to the Ar or Mr of the substance in grams (its molar mass). Using Ar: Li = 7 and O = 16:

- 1 mol of lithium is 6.02×10^{23} Li atoms, with a mass of 7 g.
- 1 mol of oxygen gas is 6.02×10^{23} O_2 molecules, with a mass of $2 \times 16 = 32$ g.
- 1 mol of lithium oxide is 6.02×10^{23} Li_2O formula units, with a mass of $(2 \times 7) + 16 = 30$ g.

Activity C3.02 — Question 2

Your explanation should include how the balanced equation shows the mole ratios.

Answer: The balancing numbers in front of the formulae give the numbers of moles that react and are formed. $4Li + O_2 \rightarrow 2Li_2O$ means that 4 mol of lithium atoms react with 1 mol of oxygen molecules to form 2 mol of lithium oxide, so the mole ratio Li : O_2 : Li_2O is 4 : 1 : 2.

Activity C3.02 — Question 3

Your explanation should include how these ratios remain constant, e.g. what would happen if you had 400 moles of Li in terms of the moles of Li_2O produced, or if you only had 0.6 moles of Li.

Answer: The ratio 4 : 1 : 2 is fixed whatever amounts are used. Every 4 mol of Li gives 2 mol of Li_2O, so the moles of Li_2O are always half the moles of Li.

- 400 mol of Li: moles of $Li_2O = 400 \times 2 \div 4 =$ **200 mol**, using $400 \div 4 = 100$ mol of O_2.
- 0.6 mol of Li: moles of $Li_2O = 0.6 \times 2 \div 4 =$ **0.3 mol**, using $0.6 \div 4 = 0.15$ mol of O_2.

Activity C3.02 — Question 4

Your explanation should include how an understanding of the number of moles allows masses to be deduced by use of the equation: number of moles = mass (g) ÷ molar mass.

Answer: Rearranging gives mass = number of moles × molar mass, so once the numbers of moles are known from the equation, the masses can be calculated. Using Ar: Li = 7 and O = 16, the molar masses are Li = 7 g/mol, O_2 = 32 g/mol and Li_2O = 30 g/mol.

- 4 mol of Li = $4 \times 7 = 28$ g; 1 mol of O_2 = 32 g; 2 mol of $Li_2O = 2 \times 30 = 60$ g. The masses balance (28 g + 32 g = 60 g), as the law of conservation of mass requires.
- Example: what mass of Li_2O is formed from 2.8 g of lithium? Moles of Li = 2.8 g ÷ 7 g/mol = 0.40 mol; moles of $Li_2O = 0.40 \div 2 = 0.20$ mol; mass of $Li_2O = 0.20$ mol × 30 g/mol = **6.0 g**.

Activity C3.02 — Question 5

You might want to explain why the number of moles in the reactants and products does not have to be the same. In this case, for example, there are five moles of reactants but two moles of products.

Answer: A chemical reaction conserves **atoms** (and therefore mass), not the number of particles. In $4Li + O_2 \rightarrow 2Li_2O$ there are 4 Li atoms and 2 O atoms on each side, and the mass is 60 g on each side. However, the atoms are regrouped: 4 separate Li atoms and 1 O_2 molecule (5 particles) become 2 Li_2O formula units, each containing 3 atoms. Fewer, larger particles are formed, so 5 mol of reactants give only 2 mol of product.

Activity C3.02 — Question 6

Having worked through this example, can you think of another balanced equation and produce a couple of questions to test your classmates' understanding of moles?

Answer:

Example equation: $2Mg + O_2 \rightarrow 2MgO$ (Ar: Mg = 24, O = 16; Mr(MgO) = 40).

- Question: How many moles of MgO are formed from 3 mol of Mg, and how many moles of O_2 are needed? Answer: the mole ratio Mg : O_2 : MgO is 2 : 1 : 2, so 3 mol of MgO is formed, using 1.5 mol of O_2.
- Question: What mass of MgO is formed when 6 g of Mg burns completely? Answer: moles of Mg = 6 g ÷ 24 g/mol = 0.25 mol, so 0.25 mol of MgO is formed; mass of MgO = 0.25 mol × 40 g/mol = 10 g.

Activity C3.02 — Peer Assessment

Having constructed a brief lesson on the mole, discuss how successful you thought it was with other members of your group, using these questions: Were you able to link the ideas suggested in the activity convincingly? Did your group think your presentation was clear and understandable? What would you change if you were to repeat the activity? Could you extend your ideas to include reactions between gases, or in solution?

Answer:

- Linking the ideas: a convincing lesson moves step by step from what a mole is (6.02×10^{23} particles, with a mass equal to Mr in grams), to the mole ratio in the equation (4 : 1 : 2), to calculating masses using moles = mass ÷ molar mass.
- Clarity: following one example all the way through, with units at every step, makes the presentation easy to understand.
- Changes: add a diagram of the "bridge" (mass → moles → ratio → moles → mass) and a short quiz at the end to check understanding.
- Extension: for gases, 1 mol of any gas occupies 24 dm^3 at r.t.p., so gas volumes react in the same ratio as moles (e.g. $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ is 1 volume + 1 volume → 2 volumes). For solutions, the concentration in g/dm^3 gives the mass of solute in a known volume (mass = concentration × volume in dm^3), and this mass can be converted into moles.

Reflection — Learning About the Mole

Think about the following questions: Did you face any problems when learning about the concept of the mole in this chapter? Are there any maths skills that you feel it would be helpful to improve?

Answer:

- Common problems: the mole is both a number of particles (6.02×10^{23}) and a mass (Mr in grams), which can be confusing at first; it is easy to forget the balancing numbers when using the mole ratio; and it is easy to mix up cm^3 and dm^3 in gas calculations.
- Maths skills worth improving: rearranging equations (the calculation triangle); working with standard form, such as 6.02×10^{23}; converting units ($cm^3 \div 1000 = dm^3$); using ratio and proportion; and rounding to sensible significant figures only at the end.

Question C3.10

Calculate the number of moles of gas there are in:

(a) 480 cm^3 of argon

(b) 48 dm^3 of carbon dioxide

(c) 1689 cm^3 of oxygen

Answer:

Taking the gases to be at r.t.p.: number of moles = volume (dm^3) ÷ 24 dm^3/mol, after converting cm^3 into dm^3 by dividing by 1000.

(a) 480 cm^3 = 480 ÷ 1000 = 0.480 dm^3; number of moles = 0.480 dm^3 ÷ 24 dm^3/mol = **0.020 mol**

(b) number of moles = 48 dm^3 ÷ 24 dm^3/mol = **2.0 mol**

(c) 1689 cm^3 = 1689 ÷ 1000 = 1.689 dm^3; number of moles = 1.689 dm^3 ÷ 24 dm^3/mol = 0.070375 = **0.0704 mol** (3 significant figures)

Question C3.11

Calculate the volume (in cm^3) of the following at r.t.p.:

(a) 1.5 moles of nitrogen

(b) 0.06 moles of ammonia

(c) 0.5 moles of chlorine

Answer:

Volume = number of moles × molar gas volume, where the molar gas volume at r.t.p. = 24 dm^3/mol = 24 000 cm^3/mol.

(a) volume = 1.5 mol × 24 000 cm^3/mol = **36 000 cm^3** (36 dm^3)

(b) volume = 0.06 mol × 24 000 cm^3/mol = **1440 cm^3**

(c) volume = 0.5 mol × 24 000 cm^3/mol = **12 000 cm^3** (12 dm^3)

Project C3.01 — Discussion Question 1

The mole and a green approach to industrial chemistry: the mole and the ratios in which chemicals react are very important in deciding the conditions used in key industrial processes such as the Haber and Contact processes. These ideas are part of the Green Chemistry approach, which is set out in 12 principles: 1 prevent waste; 2 maximise atom economy; 3 less hazardous chemicals; 4 safer chemicals and products; 5 safer solvents and reaction conditions; 6 increase energy efficiency; 7 use renewable feedstocks; 8 reduce derivatives; 9 use catalysts; 10 design degradable products; 11 analyse for pollution control; 12 accident prevention. In groups, research and discuss the principles of Green Chemistry. Why might a green approach to chemistry be important for industrial processes?

Answer:

- Industrial processes use and produce enormous quantities of chemicals, so even a small percentage of waste is a very large amount.
- A green approach reduces pollution and the risks to workers, local people and the environment from hazardous substances.
- It conserves limited raw materials and energy resources by using renewable feedstocks and energy-efficient processes.
- It saves money: less raw material is wasted, less energy is used and less waste has to be treated or disposed of.
- It helps companies meet environmental laws and public expectations, so that industry can continue in a sustainable way.

Project C3.01 — Discussion Question 2

Which of the principles do you think are the most important? Why?

Answer:

Model answer (any well-justified choice is valid):

- **Prevent waste (principle 1)** is arguably the most important: it is better not to make waste at all than to treat or clean it up afterwards, and it saves both raw materials and money.
- **Maximise atom economy (principle 2)** is closely linked: if most of the atoms in the raw materials end up in the desired product, very little is wasted.
- **Use catalysts (principle 9)** and **increase energy efficiency (principle 6)** let reactions run at lower temperatures and pressures, which saves energy and reduces the pollution from burning fuels.
- **Accident prevention (principle 12)** protects people's lives, which matters more than any economic gain.

Project C3.01 — Discussion Question 3

How do these ideas link to what you have learnt in this chapter?

Answer:

- Balanced equations show the exact mole ratios in which substances react, so chemists can calculate how much of each raw material is needed without wasting any.

- Reacting-mass calculations (using moles and Mr) predict how much product, and how much by-product or waste, a process will make; atom economy is worked out from the same balanced equation and relative formula masses.
- Knowing the limiting reactant means the amounts can be controlled so that unreacted material is not wasted, or so that a cheap reactant is used in excess to make sure an expensive one is completely used up.
- Gas volume calculations (24 dm^3 per mole at r.t.p.) are used to control the amounts of gases in processes such as the Haber process.

Project C3.01 — Podcast Question 1

Following your discussion, work in groups to prepare a podcast for the class explaining the importance of the green approach to industrial chemistry. Why is it important to have a balanced symbol equation for your reaction?

Answer:

- A balanced symbol equation shows the correct formulae of all the reactants and products, including any by-products that will have to be dealt with.
- It gives the mole ratio (the stoichiometry) of the reaction. Together with molar masses, this lets chemists calculate the exact masses (or gas volumes) of raw materials needed and the maximum mass of product that can be made.
- Using exactly the right amounts avoids wasting raw materials and energy, and it allows the atom economy of the process to be worked out and improved.
- It obeys the law of conservation of mass, so every atom can be accounted for.

Project C3.01 — Podcast Question 2

Why is it important to consider limiting reactants and those in excess in an industrial reaction?

Answer:

- The limiting reactant decides the maximum amount of product that can be made; adding more of the other reactant will not make any more product.
- Any reactant in excess is left over at the end. It is wasted unless it is separated and recycled, which costs money and energy, and it may contaminate the product or cause pollution.
- Sometimes a cheap reactant (such as oxygen from the air) is deliberately used in excess to make sure that all of a more expensive reactant is used up, as when aluminium is reacted with excess oxygen.
- Calculating the correct amounts keeps waste and costs to a minimum, which supports the Green Chemistry principles of preventing waste and maximising atom economy.

Practice Question 1

Every atom has its own atomic mass, Ar. These are added together to find the relative formula mass, Mr, of a substance. The Mr value for calcium carbonate, $CaCO_3$, is 100. Identify what mass of carbon is present in 100 g of calcium carbonate. [1]

A: 12 g

B: 60 g

C: 48 g

D: 36 g

Answer: A, 12 g. [1]

Each formula unit of $CaCO_3$ contains one carbon atom (Ar = 12), and $Mr(CaCO_3) = 40 + 12 + (3 \times 16) = 100$. So 100 g of calcium carbonate (1 mol) contains 1 mol of carbon atoms, which is 12 g: mass of carbon = $(12 \div 100) \times 100$ g = 12 g.

B (60 g) is the mass of the carbonate group, CO_3 (12 + 48); C (48 g) is the mass of the oxygen (3 × 16); D (36 g) wrongly counts three carbon atoms.

Practice Question 2

Hydrogen peroxide solution is used as a bleach and to clean oil paintings. A solution of hydrogen peroxide contains 10 g of hydrogen peroxide in every 100 cm³ of solution. Identify the concentration of the solution in g/dm³. [1]

A: 10 g/dm³

B: 20 g/dm³

C: 50 g/dm³

D: 100 g/dm³

Answer: D, 100 g/dm³. [1]

Volume = 100 cm³ ÷ 1000 = 0.100 dm³

Concentration = mass ÷ volume = 10 g ÷ 0.100 dm³ = 100 g/dm³

In other words, 1 dm³ (1000 cm³) is ten times 100 cm³, so it contains 10 × 10 g = 100 g. A (10 g/dm³) forgets to convert 100 cm³ into dm³.

Practice Question 3

Aqueous phosphoric acid reacts with magnesium carbonate to form magnesium phosphate, water and carbon dioxide gas.

(a) Phosphoric acid has the structure shown in the figure: a central phosphorus atom joined by a double bond to one oxygen atom and by single bonds to three –OH groups. State its molecular formula. [1]

(b) Give the word equation for the reaction between magnesium carbonate and phosphoric acid, including state symbols. [2]

(c) Solid magnesium carbonate has the formula $MgCO_3$. Calculate its relative formula mass. [2]

(d) Define relative formula mass. [1]

(e) 6 g of magnesium reacts to form 21 g of magnesium carbonate. Calculate the mass of magnesium carbonate formed from 1.2 g of magnesium. [2]

[Total: 8]

Answer:

(a) **H_3PO_4**: the structure contains 3 H atoms, 1 P atom and 4 O atoms (1 in P=O and 3 in the –OH groups). [1]

(b) magnesium carbonate(s) + phosphoric acid(aq) → magnesium phosphate(s) + water(l) + carbon dioxide(g) [2]

Magnesium carbonate is a solid, phosphoric acid is aqueous, water is a liquid and carbon dioxide is a gas. Magnesium phosphate is insoluble in water, so it forms as a solid. (The balanced symbol equation is $3MgCO_3(s) + 2H_3PO_4(aq) \rightarrow Mg_3(PO_4)_2(s) + 3H_2O(l) + 3CO_2(g)$.)

(c) Mr($MgCO_3$) = 24 + 12 + (3 × 16) [1] = 24 + 12 + 48 = **84** [1] (Ar: Mg = 24, C = 12, O = 16)

(d) The relative formula mass (Mr) is the sum of the relative atomic masses of all the atoms present in one formula unit of a substance. [1]

(e) 6 g of Mg forms 21 g of $MgCO_3$, so 1 g of Mg forms 21 ÷ 6 = 3.5 g of $MgCO_3$. [1]

So 1.2 g of Mg forms 1.2 × 3.5 = **4.2 g of $MgCO_3$**. [1]

Check: from the formula, 24 g of Mg forms 84 g of $MgCO_3$, the same ratio of 3.5.

Practice Question 4

When propane, C_3H_8, burns completely in air, carbon dioxide and water are formed.

(a) Give a symbol equation for the reaction. [2]

(b) Calculate the volume of carbon dioxide, measured at r.t.p., that will be formed when 100 cm³ of propane is burned. [2]

(c) 50 cm³ of methane is burned in 150 cm³ of oxygen: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. Calculate the total volume of the mixture after reaction measured at r.t.p. [3]

[Total: 7]

Answer:

(a) **$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$** [2]

Check: 3 C, 8 H and 10 O atoms on each side. With state symbols: $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$.

(b) From the equation, 1 mol of C_3H_8 gives 3 mol of CO_2. By Avogadro's law, gas volumes are in the same ratio as moles, so 1 volume of propane gives 3 volumes of carbon dioxide. [1]

Volume of CO_2 = 3 × 100 cm³ = **300 cm³** [1]

(c) From the equation, 1 volume of CH_4 reacts with 2 volumes of O_2 to give 1 volume of CO_2.

- Oxygen used = 2 × 50 cm³ = 100 cm³, so oxygen left over (in excess) = 150 − 100 = 50 cm³. [1]
- Carbon dioxide formed = 50 cm³. [1]
- At r.t.p. water is a liquid, so its volume is negligible and it is not counted in the gas volume.
- Total volume of the mixture after reaction = 50 cm³ of O_2 + 50 cm³ of CO_2 = **100 cm³** [1]

Practice Question 5

The structure of butanoic acid is shown in the figure: a chain of four carbon atoms, $CH_3–CH_2–CH_2–COOH$. The first carbon carries 3 hydrogen atoms and the next two carry 2 hydrogen atoms each; the fourth carbon is joined by a double bond to an oxygen atom and by a single bond to an –OH group.

(a) Give the molecular formula of butanoic acid. [1]

(b) Calculate the molecular mass of butanoic acid. [1]

(c) Butanoic acid dissolves in water forming butanoate and hydrogen ions: $C_3H_7COOH \rightarrow C_3H_7COO^- + H^+$. Magnesium reacts with butanoic acid solution to form a salt, magnesium butanoate, and hydrogen. Give a symbol equation for this reaction, including state symbols. [4]

(d) Butanoic acid has a molecular mass. Magnesium butanoate has a formula mass. Outline one definition that applies to both these terms. [2]

(e) Explain why different terms are used for the two quantities. [1]

[Total: 9]

Answer:

(a) **$C_4H_8O_2$** (4 C, 8 H and 2 O atoms; the formula C_3H_7COOH shows the same atoms grouped to show the structure) [1]

(b) Mr = (4 × 12) + (8 × 1) + (2 × 16) = 48 + 8 + 32 = **88** [1]

(c) **$Mg(s) + 2C_3H_7COOH(aq) \rightarrow (C_3H_7COO)_2Mg(aq) + H_2(g)$** [4]

- Magnesium forms Mg^{2+} ions and each butanoate ion is $C_3H_7COO^-$, so magnesium butanoate is $(C_3H_7COO)_2Mg$ (it can also be written $Mg(C_3H_7COO)_2$).
- Two molecules of acid are needed to provide the two H^+ ions that form one H_2 molecule. Check: 1 Mg, 8 C, 16 H and 4 O atoms on each side.
- The reaction is really between magnesium and the hydrogen ions: $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$.

(d) Both are the **sum of the relative atomic masses of all the atoms shown in the formula** of the substance [1], that is, in one molecule (molecular mass) or one formula unit (formula mass), measured on the scale on which an atom of carbon-12 has a mass of exactly 12, so they have no units [1].

(e) Butanoic acid is a **covalent** substance made of separate **molecules**, so its mass is a molecular mass. Magnesium butanoate is **ionic**: it is a giant lattice of Mg^{2+} and $C_3H_7COO^-$ ions with no separate molecules, so its formula represents a **formula unit** and its mass is a formula mass. [1]

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7. Exam Focus Priorities (Ranked by Frequency)

1. **Moles from masses** — amount (mol) = mass ÷ molar mass and its rearrangements; working out molar masses in g/mol; this appears in almost every calculation question.
2. **Writing and balancing symbol equations** — correct formulae first, then numbers in front only; state symbols are often asked for, and water formed is (l).
3. **Relative formula mass (Mr)** — adding Ar values, taking care with brackets and water of crystallisation; defining Ar, Mr and the difference between molecular mass and formula mass.
4. **Reacting masses** — by simple proportion, or by the bridge (mass → moles → mole ratio → moles → mass), always using the balancing numbers.
5. **Gas volume calculations** — 24 dm³ (24 000 cm³) per mole at r.t.p.; converting cm³ and dm³; volume ratios for reacting gases, including the gas left in excess and ignoring liquid water.
6. **Formulae of ionic compounds** — using ion charges so that the charges balance; brackets around compound ions, e.g. $(NH_4)_2SO_4$, $Mg(NO_3)_2$.
7. **Ionic equations** — splitting aqueous ionic compounds into ions, removing spectator ions; $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ and precipitation reactions.
8. **Limiting reactant and excess** — moles ÷ balancing number, smallest value is limiting; the limiting reactant fixes the amount of product.
9. **Formulae from structures** — counting atoms in diagrams to give a molecular formula; recognising ionic lattices (formula unit) versus covalent molecules.
10. **Definitions** — mole, Avogadro constant (6.02×10^{23}), molar mass, molar gas volume, r.t.p., relative atomic and formula mass.
11. **Concentration in g/dm³** — mass ÷ volume in dm³, converting cm³ by dividing by 1000.
12. **Fixed composition of compounds** — the magnesium oxide experiment: finding the mass of oxygen by difference; a straight-line graph shows a fixed ratio by mass.

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8. Quick Revision Card

- **Symbols:** one capital letter, with any second letter lower case; Latin-based symbols include Na, K, Fe, Cu, Ag, Au, Pb.
- **Elements:** metals, giant covalent elements and noble gases are written as just the symbol; simple molecules are H_2, N_2, O_2, Cl_2, P_4, S_8.
- **Covalent formulae:** valency is the group number for Groups I–IV and 8 − group number for Groups V–VII; cross over and simplify ($C_2O_4 \rightarrow CO_2$).
- **Ionic formulae:** charges must balance; put compound ions in brackets when more than one is needed: Al_2O_3, CaO, Na_2CO_3, $(NH_4)_2SO_4$, $Mg(NO_3)_2$.
- **Balancing:** never change a formula, only add numbers in front; atoms are rearranged, so mass is conserved.
- **State symbols:** (s), (l), (g), (aq); water formed in a reaction is (l).
- **Ionic equations:** cancel the spectator ions; neutralisation is $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$.
- **Ar** is measured relative to carbon-12 = 12; chlorine is 35.5 because of its isotopes; **Mr** = sum of the Ar values, with no units.
- A compound always has the same composition by mass (magnesium oxide is 60% Mg and 40% O).
- **1 mole** = 6.02×10^{23} particles (Avogadro constant, L) = Mr in grams (molar mass, g/mol).

- **moles = mass ÷ molar mass**; mass = moles × molar mass.
- **Bridge:** mass → moles → use the ratio from the equation → moles → mass.
- **Limiting reactant:** divide the moles of each reactant by its balancing number; the smallest is limiting and decides the amount of product.
- **Gases:** 1 mol of any gas = 24 dm^3 = 24 000 cm^3 at r.t.p. (25 °C, 101.3 kPa); moles = volume (dm^3) ÷ 24.
- **Reacting gases:** volumes are in the same ratio as moles ($H_2 + Cl_2 \rightarrow 2HCl$ is 1 : 1 : 2).
- **Concentration (g/dm^3)** = mass (g) ÷ volume (dm^3); 1 dm^3 = 1 litre = 1000 cm^3.